Predicting the NBA Playoffs

CMSE 202 Section 004

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Our Goal:

Create a model to accurately predict NBA playoff matchups

- Which stats are the most telling of which team is better?
- How can we use data to create a function?
- How can we test our model?

Data Set

https://www.kaggle.com/datasets/mharvnek/nba-team-stats-00-to-18

```
nba = pd.read_csv('nba_team_stats_00_to_21.csv')
nba.head()
```

	teamstatspk	TEAM	GP	W	L	WIN%	MIN	PTS	FGM	FGA	...	REB	AST	TOV	STL	BLK	BLKA	PF	PFD	+/-	SEASON
0	0	Phoenix Suns	52	42	10	0.808	48.1	112.7	42.7	89.4	...	46.1	26.5	13.3	8.6	4.3	4.0	19.3	19.3	7.8	2020-21
1	1	Golden State Warriors	53	40	13	0.755	48.2	110.9	40.4	86.5	...	46.5	27.5	15.6	9.4	4.9	4.1	20.3	17.7	8.3	2020-21
2	2	Memphis Grizzlies	55	37	18	0.673	48.3	112.7	42.7	93.4	...	48.6	25.1	13.3	10.1	6.4	6.4	19.1	19.0	4.1	2020-21
3	3	Miami Heat	54	34	20	0.630	48.5	108.7	39.3	85.7	...	44.6	25.9	14.9	7.6	3.3	4.4	20.5	20.0	4.2	2020-21
4	4	Chicago Bulls	53	33	20	0.623	48.1	111.6	41.6	87.0	...	43.0	24.5	13.0	7.2	4.6	5.2	18.8	17.8	1.7	2020-21

5 rows × 29 columns

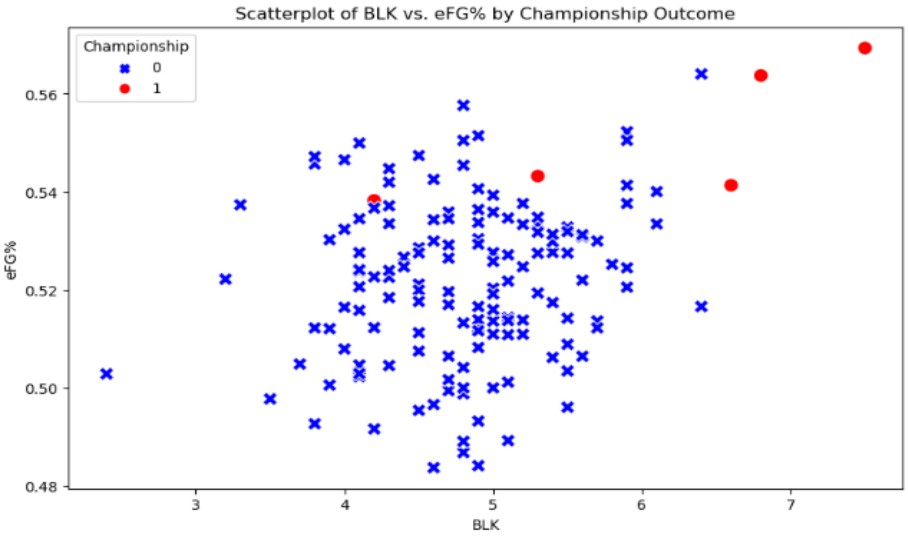
- Data found from Basketball Reference.com and NBA.com

```
nba2024.head()
```

	TEAM	WIN%	+/-	FG%	3P%	BLK
0	Boston Celtics	0.792	11.6	48.6	38.8	6.5
1	Minnesota Timberwolves	0.688	6.9	48.4	38.8	6.0
2	Denver Nuggets	0.688	4.7	49.5	37.4	5.5
3	Oklahoma City Thunder	0.675	6.4	49.8	38.7	6.6
4	LA Clippers	0.636	3.7	49.1	38.5	5.1

Examining Statistics

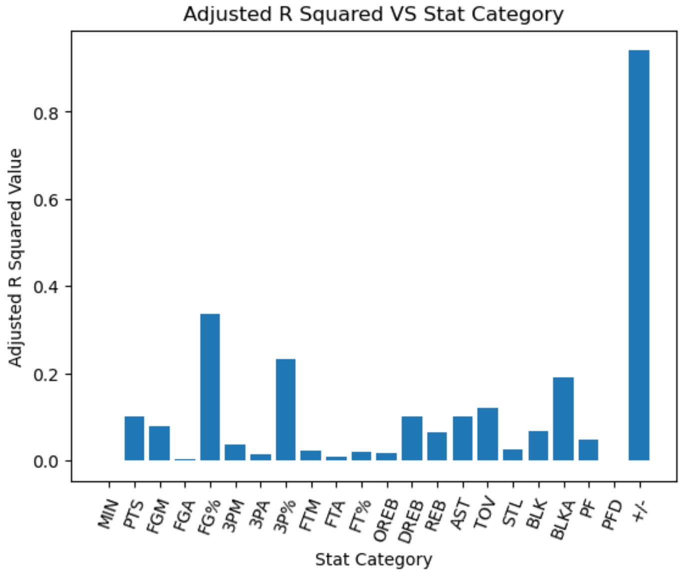
- Using a scatterplot we tested multiple variables to see which led to a championship in recent years
- Found effective field goal percentage which is an overall measure of shooting efficiency
- Examining this plotted against blocks per game for a team revealed the championship teams of the last 5 years excel in these areas
- Shows championship teams have shooting talent and good rim protection




```
last5['Championship'] = 0

champions = {
    '2020-21': 'Milwaukee Bucks',
    '2019-20': 'Los Angeles Lakers',
    '2018-19': 'Toronto Raptors',
    '2017-18': 'Golden State Warriors',
    '2016-17': 'Golden State Warriors'
}

for season, team in champions.items():
    last5.loc[(last5['SEASON'] == season) & (last5['TEAM'] == team), 'Championship'] = 1

last5['eFG%'] = (last5['FGM'] + 0.5 * last5['3PM']) / last5['FGA']

plt.figure(figsize=(10, 6))
sns.scatterplot(x='BLK', y='eFG%', data=last5, hue='Championship', style='Championship', s=100,
                palette={1: 'red', 0: 'blue'}, markers={1: 'o', 0: 'X'})
plt.title('Scatterplot of BLK vs. eFG% by Championship Outcome')
plt.xlabel('BLK')
plt.ylabel('eFG%')
plt.show()
```

OLS Model

- Used linear regression from statsmodels OLS.
- Predicting WIN%, and what variables correlate with it most.
- +/-, FG%, and 3P%.




```python
def check_rsq4(STAT):
    ''' Take in a stat and run it through an OLS model
    to determine it's correlation with the stat WIN%'''


    y = nba['WIN%']
    x = nba[STAT]

    # Adding a constant
    x = sm.add_constant(x)

    model = sm.OLS(y, x).fit()

    adj_rsquared = model.rsquared_adj

    return adj_rsquared
```

The Function

- Makes a df for the season and for both teams
- Adds up different stat categories with different weights to compute a rating for each team
- Decides which team will win based on whose rating is higher
- Decides how many games it will take (7 game series) based on how much higher the winning team's rating is.

```
def headtohead(season, team1, team2):

    df = nba[nba['SEASON'] == season]
    team1 = df[df['TEAM'] == team1]
    team2 = df[df['TEAM'] == team2]

    rating1 = ((team1['WIN%'] + team1['+/-']/5 + team1['FG%']/50 +
           team1['3P%']/50 + team1['BLK']/10).values[0])

    rating2 = ((team2['WIN%'] + team2['+/-']/5 + team2['FG%']/50 +
           team2['3P%']/50 + team2['BLK']/10).values[0])

    if rating1 > rating2:
        if 1 < rating1/rating2 < 1.25:
            print('pick', team1['TEAM'].values[0], 'in 7 games')
        if 1.25 < rating1/rating2 < 1.5:
            print('pick', team1['TEAM'].values[0], 'in 6 games')
        if 1.5 < rating1/rating2 < 1.75:
            print('pick', team1['TEAM'].values[0], 'in 5 games')
        if 1.75 < rating1/rating2:
            print('pick', team1['TEAM'].values[0], 'in 4 games')

    if rating2 > rating1:
        if 1 < rating2/rating1 < 1.25:
            print('pick', team2['TEAM'].values[0], 'in 7 games')
        if 1.25 < rating2/rating1 < 1.5:
            print('pick', team2['TEAM'].values[0], 'in 6 games')
        if 1.5 < rating2/rating1 < 1.75:
            print('pick', team2['TEAM'].values[0], 'in 5 games')
        if 1.75 < rating2/rating1:
            print('pick', team2['TEAM'].values[0], 'in 4 games')
```

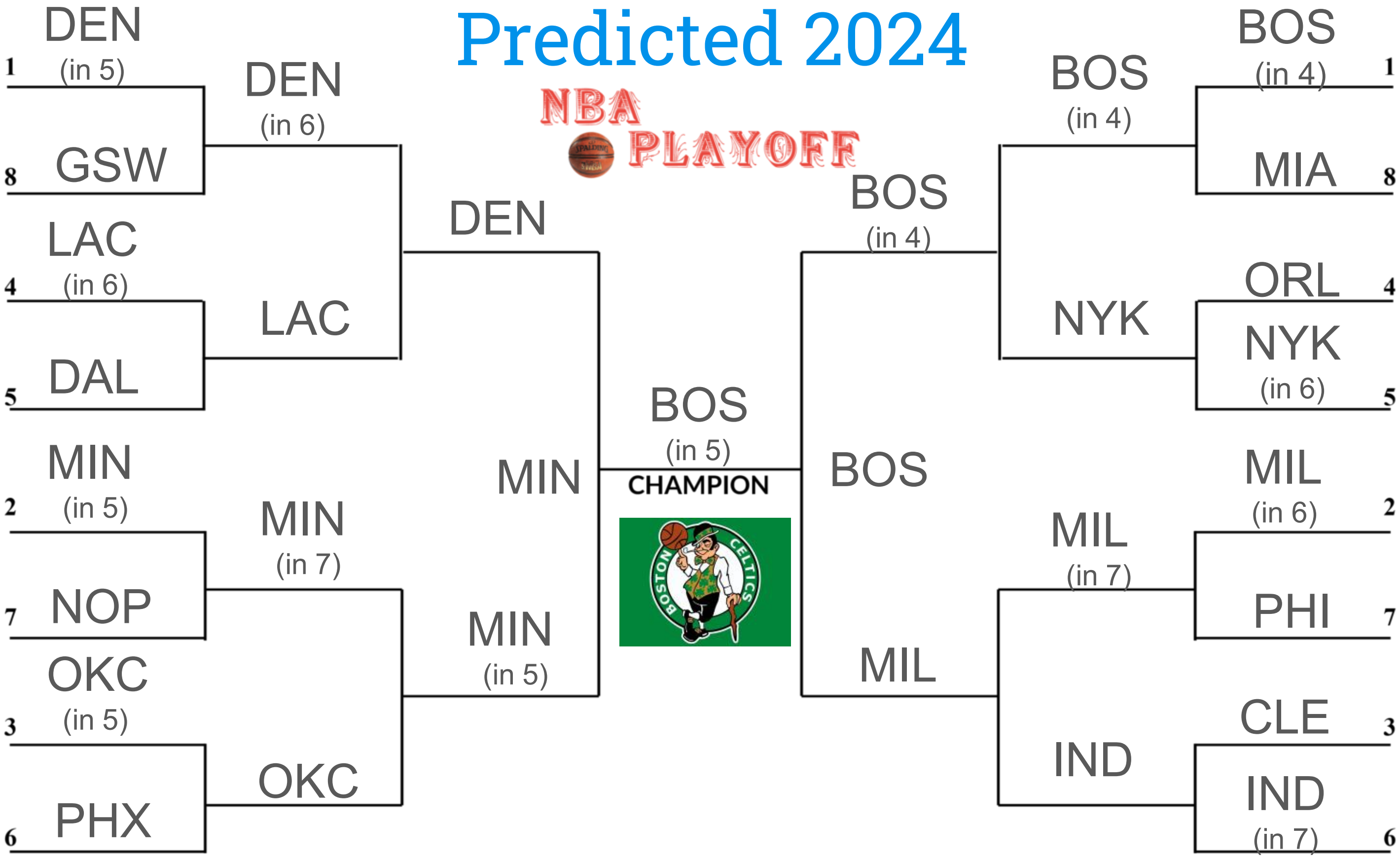
```
headtohead("2014-15", 'Golden State Warriors', 'New Orleans Pelicans')
```

```
pick Golden State Warriors in 5 games
```

2015

Correctly Predicted 11/15 games

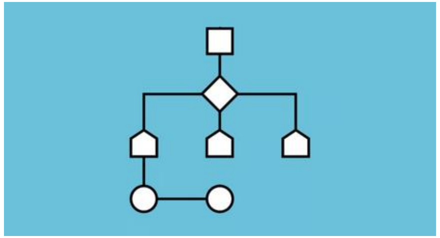
Predicted 2024



*Standings and statistics as of 4/7/2024

Difficulties or Complications

- Choosing which stats to use in the function
- How to weigh the rating system
- Not having a complete set of data for this season
- Working with parameters for determining the length of the series



Questions?

Sources

https://www.kaggle.com/datasets/mharvnek/nba-team-stats-00-to-18

https://www.espn.com/nba/bracket/_/year/2015

Slide theme from:

https://www.slidescarnival.com/cordelia-free-presentation-template/216#

Image sources:

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